BOARDGUARD · VISUAL INSPECTION REPORT

BG-20261026-4D2ECF

Created 2026-10-26 17:38:04 UTC · mode inspection · board region REGION-B · Fine-pitch controller IC · scenario S02-bridge-glare · OpenCV 5.0.0 · processed on local

> **COMPLETED** — 1 finding(s) confirmed visually by technician; 1 dismissed; electrical verification pending.

Observations

ID	ROLE	ORIGINAL SIZE	QUALITY	BLUR METRIC	GLARE FRACTION	REASONS	REGISTRATION
OBS-001	initial	1597×1600	ACCEPT	0.3201	0.0159	–	–
OBS-002	recapture	1597×1600	ACCEPT	0.32	0.0141	–	REGISTERED · 3923 inliers



OBS-001 · initial · TQFP-144 under ring light (specular glare on right lead row) · SYNTHETIC: specular highlight (Gaussian, r=50 px) added with OpenCV to a real photograph to emulate ring-light glare.



OBS-002 · recapture · TQFP-144 recaptured with diffuse light · Recorded photograph (real TQFP-144, 0.5 mm pitch, no glare).

Findings (suspected visual anomalies)

ID	TYPE	STATUS	PRIORITY	EVIDENCE SCORE*	HISTORY	TECHNICIAN	RECOMMENDED CHECK
C01	suspected solder bridge	TECHNICIAN_DISMISSED	LOW	0.787	DETECTED → NOT_REPRODUCED	dismiss — Not visible under diffuse light; reflection artefact.	Continuity / resistance check between the adjacent pins with a multimeter (power off); inspect under a second lighting angle.
C02	suspected solder bridge	TECHNICIAN_CONFIRMED	HIGH	0.661	DETECTED → PERSISTED	confirm — Fibre across adjacent leads; remove	Continuity / resistance check between the adjacent pins with

						and check continuity.	a multimeter (power off); inspect under a second lighting angle.

* Heuristic evidence score from image measurements; not a calibrated probability.

Technician actions

- 2026-10-26 17:38:09 UTC — candidate_decision · C01 — Not visible under diffuse light; reflection artefact.
- 2026-10-26 17:38:09 UTC — candidate_decision · C02 — Fibre across adjacent leads; remove and check continuity.
- 2026-10-26 17:38:09 UTC — finalize — Visual inspection closed; electrical check scheduled.

Agent trace

#	TIME (UTC)	ACTOR	EVENT	TOOL	RULE	NEXT STATE	MS
1	17:38:04.311	system	Inspection session created			CREATED	
2	17:38:04.356	system	OBS-001 received (initial)			CAPTURE_RECEIVED	
3	17:38:04.356	agent	Quality assessment started			QUALITY_ASSESSMENT	
4	17:38:04.726	agent	Assess capture quality of OBS-001	assess_image_quality		QUALITY_ASSESSMENT	370.7
5	17:38:04.726	agent	Quality gate passed (ACCEPT)		R-Q3	INSPECTION_REQUIRED	
6	17:38:06.626	agent	Localise suspected anomalies in OBS-001	detect_visual_candidates		INSPECTION_REQUIRED	1899.6
7	17:38:06.626	agent	2 suspected anomalies localised			CANDIDATE_DETECTED	
8	17:38:06.628	agent	Check local evidence for C01	analyze_candidate_region		CANDIDATE_DETECTED	1.4
9	17:38:06.629	agent	Check local evidence for C02	analyze_candidate_region		CANDIDATE_DETECTED	1.2
10	17:38:06.629	agent	Evidence insufficient for a visual decision		R-E1	EVIDENCE_INSUFFICIENT	
11	17:38:06.629	agent	Recapture requested	request_recapture	R-E1	RECAPTURE_REQUESTED	

12	17:38:06.665	system	OBS-002 received (recapture)			SECOND_OBSERVATION_RECEIVED	
13	17:38:06.665	agent	Quality assessment started			QUALITY_ASSESSMENT	
14	17:38:07.048	agent	Assess capture quality of OBS-002	assess_image_quality		QUALITY_ASSESSMENT	382.0
15	17:38:07.048	agent	Quality gate passed (ACCEPT)		R-Q3	INSPECTION_REQUIRED	
16	17:38:08.789	agent	Localise suspected anomalies in OBS-002	detect_visual_candidates		INSPECTION_REQUIRED	1742.0
17	17:38:09.253	agent	Register OBS-002 to OBS-001	register_image_pair		INSPECTION_REQUIRED	464.1
18	17:38:09.256	agent	Compare candidates across OBS-001 and OBS-002	compare_observations		INSPECTION_REQUIRED	2.7
19	17:38:09.256	agent	Multi-view comparison complete			COMPARISON_COMPLETE	
20	17:38:09.256	agent	Reinspection changed the assessment		R-M2	COMPARISON_COMPLETE	
21	17:38:09.256	agent	Technician review requested	request_technician_review	R-M2	TECHNICIAN_REVIEW_REQUIRED	
22	17:38:09.297	technician	C01 dismissed by technician	record_verification	R-H1	TECHNICIAN_REVIEW_REQUIRED	
23	17:38:09.313	technician	C02 confirmed by technician	record_verification	R-H1	TECHNICIAN_REVIEW_REQUIRED	
24	17:38:09.313	agent	Review complete - verification pending			VERIFICATION_PENDING	
25	17:38:09.327	technician	Inspection closed by technician			VERIFICATION_PENDING	
26	17:38:09.327	agent	Inspection completed			COMPLETED	

Known limitations

- BoardGuard is a visual inspection aid. It does not measure electrical continuity, resistance or function; a photograph cannot prove or exclude a short or open circuit.
- Candidates are suspected visual anomalies localised by classical OpenCV image analysis with documented, hand-calibrated thresholds. Evidence scores are heuristic and are not calibrated probabilities.
- Thresholds were calibrated on a small set of public PCB photographs and controlled degradations of them, not on a validated corpus of repair cases.
- Software zoom is not optical magnification; lighting guidance must be carried out physically by the technician.
- A visible change between before and after captures documents a visual difference only. It does not establish that a repair succeeded.

Generated 2026-10-26 17:38:09 UTC by BoardGuard. No technician identity is recorded in the demo; actions are attributed to an unauthenticated user.